hw4

Yifeng Huang

This is the link to the repo: https://github.com/YifengHuang-Bob/Homework4

Problem 1 - Tidyverse: New Zealand

(a) Total Votes by Election Year and Voting Type

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
nzge <- readRDS("data/nzge.rds")
```

```
vote_counts <- nzge %>%
  group_by(election_year, voting_type) %>%
  summarise(total_votes = sum(votes), .groups = "drop") %>%
  arrange(desc(total_votes))

print(vote_counts, n = Inf)
```

```
# A tibble: 10 × 3
   election_year voting_type total_votes
           <dbl> <chr>             <dbl>
 1          2014 Party           2416479
```

```
 2          2014 Candidate       2375493
 3          2008 Party           2356536
 4          2008 Candidate       2325598
 5          2005 Party           2286190
 6          2005 Candidate       2260670
 7          2011 Party           2257336
 8          2011 Candidate       2225766
 9          2002 Party           2040248
10          2002 Candidate       2022115
```

(b) Candidate Vote Proportions by Party in 2014

```
candidate_votes_2014 <- nzge %>%
  filter(election_year == 2014, voting_type == "Candidate") %>%
  group_by(party) %>%
  summarise(total_votes = sum(votes), .groups = "drop") %>%
  mutate(percent_vote = 100 * total_votes / sum(total_votes)) %>%
  arrange(desc(percent_vote)) %>%
  mutate(percent_vote = round(percent_vote, 2))

print(candidate_votes_2014, n = Inf)
```

```
# A tibble: 25 × 3
   party                            total_votes percent_vote
   <chr>                                  <dbl>        <dbl>
 1 National Party                       1081787        45.5
 2 Labour Party                          801287        33.7
 3 Green Party                           165718         6.98
 4 Conservative Party                     81075         3.41
 5 New Zealand First Party                73384         3.09
 6 Maori Party                            42108         1.77
 7 MANA Movement                          32333         1.36
 8 Informal Candidate Votes               27886         1.17
 9 ACT New Zealand                        27778         1.17
10 United Future                          14722         0.62
11 Aotearoa Legalise Cannabis Party        4936         0.21
12 Internet Party                          4848         0.2
13 Democrats for Social Credit             4647         0.2
14 Ban1080                                 4448         0.19
15 Independent                             3864         0.16
16 NZ Independent Coalition                1929         0.08
```

```
17 Focus New Zealand                        1797         0.08
18 Money Free Party                          391         0.02
19 Communist League                          135         0.01
20 Climate Party                             116         0
21 Human Rights Party                         76         0
22 The Expats                                 70         0
23 Alliance                                   59         0
24 NZ Economic Euthenics Party                51         0
25 Patriotic Revolutionary Front              48         0
```

(c) Winning Parties by Election Year and Voting Type

```
winning_parties <- nzge %>%
  group_by(election_year, voting_type, party) %>%
  summarise(total_votes = sum(votes), .groups = "drop") %>%
  group_by(election_year, voting_type) %>%
  slice_max(order_by = total_votes, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  select(election_year, voting_type, party) %>%
  pivot_wider(names_from = voting_type, values_from = party) %>%
  arrange(election_year)

print(winning_parties, n = Inf)
```

```
# A tibble: 5 × 3
  election_year Candidate      Party
          <dbl> <chr>          <chr>
1          2002 Labour Party   Labour Party
2          2005 National Party Labour Party
3          2008 National Party National Party
4          2011 National Party National Party
5          2014 National Party National Party
```

Problem 2 - Tidyverse: Tennis

(a) Number of Tournaments in 2019

```
atp2019 <- read_csv("data/atp_matches_2019.csv", show_col_types = FALSE)
```

```r
tournament_count <- atp2019 %>%
  summarise(n_tournaments = n_distinct(tourney_id))

print(tournament_count)
```

```
# A tibble: 1 × 1
  n_tournaments
          <int>
1           128
```

There were **128 tournaments** in the 2019 ATP Matches dataset, determined by counting the distinct tournament IDs.

(b) Players Winning Multiple Tournaments

```r
multiple_title_winners <- atp2019 %>%
  filter(round == "F") %>%
  group_by(winner_id, winner_name) %>%
  summarise(tournaments_won = n_distinct(tourney_id), .groups = "drop") %>%
  filter(tournaments_won > 1) %>%
  arrange(desc(tournaments_won), winner_name)

print(multiple_title_winners, n = Inf)
```

```
# A tibble: 12 × 3
   winner_id winner_name        tournaments_won
       <dbl> <chr>                        <int>
 1    106233 Dominic Thiem                    5
 2    104925 Novak Djokovic                   5
 3    106421 Daniil Medvedev                  4
 4    104745 Rafael Nadal                     4
 5    103819 Roger Federer                    4
 6    200282 Alex De Minaur                   3
 7    126774 Stefanos Tsitsipas               3
 8    105332 Benoit Paire                     2
 9    106426 Cristian Garin                   2
10    104542 Jo-Wilfried Tsonga               2
11    126610 Matteo Berrettini                2
12    106401 Nick Kyrgios                     2
```

```r
multiple_title_winners %>%
  summarise(
    n_players = n(),
    max_tournaments = max(tournaments_won)
  ) %>%
  print()
```

```
# A tibble: 1 × 2
  n_players max_tournaments
      <int>           <int>
1        12               5
```

Yes. In 2019, **12 players** won more than one tournament. **Dominic Thiem and Novak Djokovic** tied for the most tournament titles, each winning **5 tournaments**.

(c) Comparing Aces Between Winners and Losers

```r
ace_comparison <- atp2019 %>%
  filter(!is.na(w_ace), !is.na(l_ace)) %>%
  mutate(ace_diff = w_ace - l_ace) %>%
  summarise(
    n_matches = n(),
    mean_winner = mean(w_ace),
    mean_loser = mean(l_ace),
    mean_diff = mean(ace_diff),
    se_diff = sd(ace_diff) / sqrt(n())
  ) %>%
  mutate(
    lower_ci = mean_diff - 1.96 * se_diff,
    upper_ci = mean_diff + 1.96 * se_diff
  ) %>%
  select(-se_diff)

print(ace_comparison, width = Inf)
```

```
# A tibble: 1 × 6
  n_matches mean_winner mean_loser mean_diff lower_ci upper_ci
      <int>       <dbl>      <dbl>     <dbl>    <dbl>    <dbl>
1      2694        7.50       5.79      1.70     1.43     1.97
```

Yes. Based on 2,694 matches with complete ace records, winners averaged **7.50 aces** per match, compared with **5.79 aces** for losers. The mean paired difference was approximately **1.70 aces** (the reported means and mean difference were calculated using unrounded values and rounded separately for display), with an approximate 95% confidence interval of **[1.43, 1.97]**.

Since the entire confidence interval is above zero, there is statistical evidence that winners tend to have more aces than losers.

(d) Players with the Highest Win Rate

```
player_results <- bind_rows(
  atp2019 %>%
    transmute(player_id = winner_id, player_name = winner_name, won = 1),
  atp2019 %>%
    transmute(player_id = loser_id, player_name = loser_name, won = 0)
)

top_win_rate <- player_results %>%
  group_by(player_id, player_name) %>%
  summarise(
    matches_played = n(),
    matches_won = sum(won),
    .groups = "drop"
  ) %>%
  filter(matches_played >= 5) %>%
  mutate(win_rate = matches_won / matches_played) %>%
  slice_max(order_by = win_rate, n = 1, with_ties = TRUE) %>%
  arrange(player_name)

print(top_win_rate, n = Inf)
```

```
# A tibble: 1 × 5
  player_id player_name  matches_played matches_won win_rate
      <dbl> <chr>                <int>       <dbl>    <dbl>
1    104745 Rafael Nadal            69          60    0.870
```

Rafael Nadal had the highest win rate among players who participated in at least 5 matches. He won **60 out of 69 matches**, achieving a win rate of approximately **87.0%**.